

Statistics and Data Science in the Wild

A True Story

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Act I — The Reality Check

Why Are You In This Room?

Take 30 seconds.

*Try to remember why you decided
to study statistics or data science.*

*I wanted a job that would NOT require me
to engage much with people.*

Boy, was I wrong!

Because it is entirely about human psychology.

And I actually like engaging with people in a statistical and data science context far more than I realized at the time.

The Math Is the Easy Part

You spent years learning the hard part of statistics.

In the industry, that turns out to be the easy part.

The techniques you have mastered are **necessary but not sufficient**. Every data scientist around you has them too. They are the entry ticket, not the differentiator.

{The degree opens the door. Everything else in this talk is about what you do once you are in the room.}

What is Actually Hard (To Me):

- Getting a room of sceptical non-statisticians to trust your result
- Knowing which question to answer when nobody has stated it clearly
- Staying useful as the tools, the field, and the organisation change around you
- Growing as a person fast enough to grow as a professional

{A brilliant model nobody uses is worse than a decent model everyone trusts.}

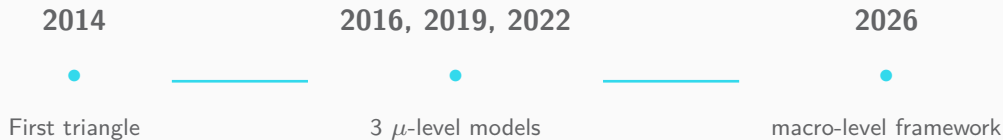
Background

- Bioengineering → Statistics → PhD Biostatistics → AI, all @ KU Leuven
- Senior Data Scientist, Ageas Re (Brussels), VSC @ UAntwerp
- 4 insurance companies over 10 years
- 15+ publications, 60+ conference talks, former Invited Lecturer
- 10+ R packages, 4 CRAN, adopted Python

Where I work: Ageas Re

- Reinsurance — internal and external — across the globe
- ~50 people — little bureaucracy
- Highly international: EN is the norm, with sprinkles of FR and NL
- High visibility and real ownership
- *Small company, hard problems: perfect blend*

A War Story: The Lost 'Decade' (1/2)



{**The problem:** insurance companies aggregate individual claims into a 10×10 summary table and run a century-old method on it. This throws away 95% of the information. The heterogeneity between claims is treated as noise rather than signal.}

A War Story: The Lost 'Decade' (2/2)

What they said

- 'The model is a black box'
- 'Excel works fine'
- 'What if the regulator asks?'

What actually worked

- Patience
- Finding internal allies
- Showing real data, not simulations

{The key unlock: numbers beat arguments. 'I have a better answer' and 'the organisation is using the better answer' are separated by years of work that has nothing to do with statistics.}

What I Would Tell My Former (2014) Self

1. **Patience is a technical skill.** The world catches up to good ideas, but not on your timeline.
2. **Find the one person who gets it.** Make them look good. That person becomes your internal champion.
3. **Regulators are sometimes your best allies.** Institutions move when they must, not when they should.
4. **The heterogeneity is the signal.** True not just in reserving, but as a general principle: *the data structure induces the geometry, which determines the optimal inferential procedure.*

Act II — How the World Actually Works

Every Data Scientist is a Consultant (1/2)

Your job title will say 'Data Scientist', 'ML/AI Engineer', or 'Data Analyst'. It does not matter. The moment someone comes to you with a problem and expects an answer, **you are consulting.**

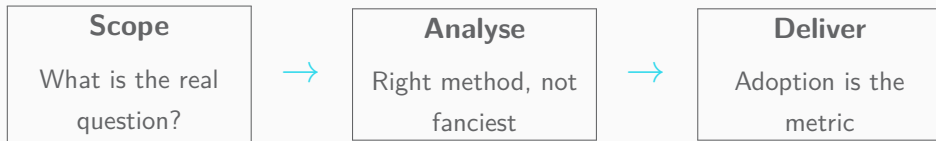
External consultant

- Hired for a specific engagement
- Authority comes from the contract
- Optimises for the deliverable
- Leaves when the project ends

Internal consultant (you)

- Embedded in the organisation
- Authority comes from trust and track record
- Optimises for *adoption and impact*
- Stays for the consequences

Every Data Scientist is a Consultant (2/2)



{The “soft” phases — scoping and delivery — are almost always harder than the analysis.}

Scoping and Translation (1/2)

The client asks for X Your first job is figuring out whether X is the right question.

<i>They say</i>	<i>They mean</i>
'Improve accuracy'	'The regulator is nervous'
'Need a dashboard'	'I look bad in meetings'
'Automate this'	'I hate doing it monthly'
'Is this significant?'	'Give me permission to act'

{ Ask not *'what do you want?'* but *'what happens if this doesn't work?'* }

Scoping and Translation (2/2)

Business → Statistics

‘How much do we owe people who haven’t asked yet?’



‘IBNR tail of an inhomogeneous marked point process under right-censoring’

Statistics → Business

‘NB-GLM with Poisson–Gamma micro-level derivation identifying κ ’



‘We can now measure precisely how much worse some years are than others.’

{Failing in either direction means the project fails.}

The Psychology of Stakeholders (1/2)

Statistical consulting is applied psychology.

The CFO

Thinks in scenarios. Needs to defend to the board.

Give: one number, a range, one action.

The actuary

Thinks in distributions. Needs regulatory cover.

Give: method, assumptions, sensitivity.

The IT team

Thinks in systems. Resists bespoke solutions.

Give: standard formats, documentation.

The sceptic

Thinks: 'this threatens something I own'.

Responds to being heard, not overruled.

Give: credit, involvement, genuine engagement.

The Psychology of Stakeholders (2/2)

When someone says 'I don't trust the model' they usually mean: *I don't understand it and I'm afraid to say so*, or *I don't trust **you** yet*. Neither is solved by better mathematics.

{Never make a stakeholder feel stupid. If they misunderstood, the failure is yours. Always give people a graceful exit.}

The Power of Humour and Human Connection (1/2)

People lower their defences when they are laughing.

- There is a default image of a data scientist in most organisations: theoretical, introverted, socially unaware, afraid of the room. **You will walk into that stereotype on day one.** Disrupt it early and it never reforms.
- You do not need to be an extrovert. You need to show that you are a **human being** — curious about the other person, capable of a joke, able to say “I don’t know” without freezing.

{The bar is not high. A data scientist who makes eye contact and asks about your weekend is already memorable.}

The Power of Humour and Human Connection (2/2)

- **I hate networking. I goof around instead** — jokes, genuine curiosity, interest in what the other person is doing. That is not a strategy. It is just being human. It works better than any strategy.
- **Put some skin in the game.** Real curiosity, real generosity, real interest in the other person. The coffee corner visit that feels like small talk is infrastructure — but only if it is genuine. People can tell the difference between someone who is performing warmth and someone who actually gives a damn.
- **The coffee corner is infrastructure.** More careers have been built at the coffee machine than in the meeting room. Someone who vouches for you spontaneously is worth more than ten lines on a CV.

How Projects Are Born and How They Die (1/2)

You need to understand **how projects come to be** and how decisions are made — because that varies enormously by company size.

Small (<100)

- Decisions by 1–2 people, informally
- Hallway conversations beat slide decks
- Speed over process
- *Get in the room*

Mid-size (100–1000)

- Committees, sign-off chains
- The slide deck *is* the decision
- Allies at multiple levels matter
- *Work the process*

Large (>1000)

- Distributed, slow, political
- Budget cycles dominate
- One champion is not enough
- *Find the network*

How Projects Are Born and How They Die (2/2)

Projects are not born technical. They are born political — someone had a problem, made a case, got budget. **Be the person who ships.** Technical brilliance that misses deadlines becomes invisible.

POC heroes are plenty. The world does not need another impressive demo that never reaches production. The real work starts after the applause. Stay for it.

{Project management is not beneath you. It is how your work reaches the world.}

Decisions Are Not Logical Processes (1/2)

We are often put on the most sensitive and strategic projects decided by people who understand very little of what we actually do.

- Very few people fully understand what you have done. What they evaluate instead is **whether they trust you, whether they like you, and whether the result feels right**. That is not irrational — it is the only instrument available to them.
- Senior management selects who runs a critical project based on **signals, not evidence**. They cannot read your code or audit your model. What they see is how you present yourself, how others speak about you, and whether you have delivered before.

{Build the reputation first. The right projects follow. Waiting for the right project to build the reputation is the wrong order.}

Decisions Are Not Logical Processes (2/2)

- **Your actual technical capability is only one part of what determines which projects you get.** The rest is visibility, track record, and the confidence you project in rooms where decisions are made.
- **Respect non-technical profiles.** The project manager keeping seventeen stakeholders aligned, the communicator who explains a complex product in one sentence — these are serious skills, not lesser because they are not quantitative.
- **The emotional layer is not noise.** A technically correct recommendation that ignores how people feel — threatened, excluded, not consulted — will lose to a weaker one that people feel ownership of.

{The most dangerous assumption a data scientist can make is that a good analysis speaks for itself.}

Data Scientists Are Marathon Runners

- Long projects involving change are friction-heavy. **Delivery goes far beyond the coding job** — even when you are not the manager. You absorb uncertainty, navigate resistance, and hold steady while the organisation shifts.
- **Control the controllables.** You cannot control the politics, the reshuffles, or the colleagues who resist. You can control the quality of your work, how you show up in hard conversations, and how you treat people under pressure. Let the rest go — deliberately, and repeatedly.

It's a Small World After All (1/2)

*The Belgian actuarial and statistics community
has approximately 300 people in it.*

You will meet the same people, over and over, for 30 years.

- People judge you **as much by how you treat them as by what you produce**. Your technical output is visible to a handful of people. How you treat people is visible to everyone.
- Criticise **methods**, not **people**. Leave every organisation in a way that people remember with respect. Be generous with credit, especially to people with less power than you.

{I have watched technically brilliant people fail because they collected enemies faster than they collected results.}

It's a Small World After All (2/2)

- **Say what you do and do what you say.** Reliability is rarer than intelligence. “I don't know, but I'll find out by Thursday” builds more trust than a confident wrong answer.
- **You can run but you cannot hide.** Inauthenticity is picked up faster than you think, and once trust is damaged it is nearly impossible to rebuild. Trust is the only currency that actually matters in this career. Do not gamble with it (or gamble in general :D).
- **When you are wrong, own it early and completely.** Defending a wrong position for status reasons is career poison. This is also good marital advice (for a man ;-))

Act III — Building a Career Deliberately

The Field Has Changed — So Has the Game (1/2)

When I started, data science barely had a name. The bar was low, the competition was thin. **That window is closed.**

What you won't have

- The novelty premium
- Prestige that comes cheaply — still real, but harder to earn
- The amateurish years — data governance and engineering standards are now baseline

What you won't have to endure

- Explaining what a data scientist is
- Building infrastructure from scratch every time
- Being the only person who has heard of cross-validation

The Field Has Changed — So Has the Game (2/2)

A good professional still gets picked up fast. Strong generalists will always have a job. **Be ready to change your mind about everything** — what feels important today may be irrelevant in ten years. Your psychological and emotional evolution amplifies your intellectual growth. The data scientists I most admire at 50 are not the ones who knew the most at 30. They are the ones who kept changing.

{Data science is living through what IT experienced in the mid-nineties: the hype is dying down and the domain is showing its true colours. That is where serious work begins.}

So Who You Wanna Be? (1/2)

Data science is an umbrella term. HR often doesn't know who they're looking for. The job ad may describe an impossibility — PhD preferred, expert in ML and cloud infrastructure and statistical modelling and business strategy, for a junior salary.

Operational / Analytical

Closest to the business.
Domain knowledge is the core asset. Dashboards, reporting, modelling for decisions.

ML Engineer / Systems

Closest to the software.
Engineering rigour is the core asset. Models in production, MLOps, scalability.

Research-oriented

Closest to the methodology.
Depth and originality are the core asset. Novel methods, publications, frameworks.

So Who You Wanna Be? (2/2)

Then comes the fork: **technical expert track** (deep mastery, career defined by problems solved) or **management track** (broad leverage, career defined by teams built). Very few navigate both successfully. *Confusing them is the most common mid-career trap.*

{Read the subtleties in job ads. Ask what a typical week looks like and be respectfully blunt about what you want. Put them under the microscope — you are evaluating them too.}

To PhD or Not to PhD? (1/2)

You are at the point where this decision is real. Here is my honest take after 20 years in industry with a PhD.

Do it if...

- You want to stay connected to academia
- You are passionate about a methodological question and can sit with it for four years
- You like to write and think in depth

Think twice if...

- The only goal is a higher starting salary
- You want breadth and variety immediately

To PhD or Not to PhD? (2/2)

The argument nobody tells you

A PhD changes your **framing**. You become capable of methodological heavy lifting that most industry data scientists cannot do. That asymmetry is permanent.

In my career, roughly **75% of my time** has been spent on research projects — for the industry. Without the PhD, I would have been put on operational work almost exclusively.

{The field is more competitive now. A PhD as a differentiator is more critical, not less — it is where serious work begins.}

Start With a Vision, Define Your Tagline (1/2)

The market is competitive. Everyone has similar training. The ones who stand out have a clear identity — a domain, a method, a perspective — that makes them *the* person for a specific thing.

This is not arrogance. It is clarity. Prepare for competition by making it **irrelevant**: be the obvious choice for *your* thing.

Be curious and open when young, because you don't yet know what you love. Be selective and ruthless once you do. *Most people make the transition too late.*

{Try to figure out who you want to be — early.}

Start With a Vision, Define Your Tagline (2/2)

Define your tagline before others define it for you

People will have a one-line description of you in their head. It is what they say when your name comes up in a room you are not in.

'The person who ships things'

'Makes complex things understandable'

'Smart but impossible to work with'

'Always interesting, never finishes'

{This tagline will normally evolve over time, for yourself and the others.}

Find Your Environment and Create It (1/2)

- **Learning what environment you thrive in is serious self-knowledge, not self-indulgence.** Do you think best alone or in conversation? With tight deadlines or long horizons? Once you know, stop apologising for it and start engineering it.
- **Neurodivergence is more common in this field than in the general population.** The same wiring that makes school difficult often produces unusual pattern recognition, obsessive depth on hard problems, and the ability to hold complex structures in working memory. Know your profile. Do not mask more than necessary.

Find Your Environment and Create It (2/2)

- **Find complementary people, not clones.** A team of five people who all think like you is not five times as strong — it has one blind spot, five times over. The best collaborations I have had felt slightly uncomfortable at first, because the other person thought differently enough to challenge me.
- **Know your limits and protect your energy.** Every career has energy vampires — activities that drain you without returning value. Saying no, clearly and kindly, is a professional skill. You cannot pour from an empty cup. Take care of your body. This is a demanding job and the cup empties faster than people admit.

Be Authentic, Honest, and Respectful (1/2)

After everything in this talk — the strategy, the positioning, the psychology, the career tracks — this is the thing I would put first if I were starting again.

- **Authentic:** show up as yourself, not as a performance of what a senior data scientist should look like. People sense the gap between a person and their persona, and they trust the person who has closed it.
- **Honest:** say what you actually think, including when it is inconvenient. Diplomatic honesty is a skill worth developing. Dishonest diplomacy is corrosive — to relationships, to your integrity, and eventually to your results.
- **Respectful:** every person in the room deserves to be treated as a full human being with their own expertise and pressures. Accept that some colleagues will be smarter than you. The people who react with curiosity rather than insecurity gain enormously. **Nobody beats you on all dimensions simultaneously.**

Be Authentic, Honest, and Respectful (2/2)

One more thing. All of this — the stakeholder management, the psychology, the reputation building — is not manipulation. It is the recognition that **you simply cannot do this work without other people**. The sooner you internalise that, the lighter it feels.

And here is the part nobody warns you about: **you will meet some genuinely remarkable people in this career**. Curious, deep, slightly unusual people who see patterns others miss and care about getting things right. People like yourself. And for perhaps the first time in your life, *you will be the norm*. The standard is set by your people.

{This combination is rarer than technical skill. And it compounds the same way.}

What This Career Actually Looks Like (1/2)

- You will spend less time on statistics than you expect — and more time on human dynamics
- The math gets you in the room; everything else determines whether you get to stay
- Domain knowledge and relationships will outlast every specific method you learn here
- People judge you as much by how you treat them as by what you produce
- The colleagues who thrive over 30 years stay curious, stay honest, stay kind — and keep changing

What This Career Actually Looks Like (2/2)

*“The data structure induces the geometry,
which determines the optimal inferential procedure.”*

The consulting version:

Understand the people before you reach for the method.

Thank you

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Your Turn

Your Turn (1/5)

You have just heard 90 minutes about what this work actually requires.

Now prove it.

Your Turn (3/5)

Higher management in an insurance company wants to identify unprofitable customers in their current portfolio in order to take appropriate action at renewal.

To support this analysis, you will receive two non-life insurance data sets — one for claim frequency, one for claim severity.

Both data sets concern the same portfolio and share a common set of covariates, but cannot be directly linked at the individual policyholder level.

Your Turn (4/5)

- There is no prescribed method.
- There is no single correct answer.
- You will be evaluated on whether you asked the right questions, made defensible choices, and can explain your reasoning clearly.

Your Turn (5/5)

